NAME: Jeevitha.S
PROJECT TOPIC:Food Delivery Management System

Assignment 7: Sample Data Insertion

DBMS Coursework | Real-World Data Simulation & Constraint Compliance

Objective

To insert realistic sample records into all eight tables of the Food Delivery Management System while satisfying the primary-key, foreign-key, UNIQUE and CHECK constraints.

1. Overview of Sample Data Ingestion Strategy

To validate functional requirements, multi-table queries, joins, and integrity rules, realistic sample data is populated across all 8 tables. Data insertion strictly adheres to foreign key dependency order: Parent tables (`Customer`, `Restaurant`, `Delivery_Partner`) are populated first, followed by child dependent tables (`Menu_Item`, `Order_Table`, `Order_Item`, `Payment`, `Delivery`).

2. SQL Data Insertion Script (Parent Entities)

```
-- 1. POPULATING CUSTOMER TABLE (5 Records)
INSERT INTO Customer (Customer_ID, Name, Phone, Email, Address) VALUES
(1, 'Aarav Sharma', '9876543210', 'aarav.sharma@gmail.com', '102 MG Road, Indiranagar, Bengaluru'),
(2, 'Priya Patel', '9812345678', 'priya.patel@yahoo.com', '45 Koramangala 5th Block, Bengaluru'),
(3, 'Rohan Verma', '9765432109', 'rohan.v@outlook.com', '12 Resident Road, Ashok Nagar, Bengaluru'),
(4, 'Sneha Reddy', '9654321098', 'sneha.reddy@gmail.com', '88 HSR Layout Sector 3, Bengaluru'),
(5, 'Vikram Malhotra', '9543210987', 'vikram.m@gmail.com', '501 Whitefield Main Rd, Bengaluru');

-- 2. POPULATING RESTAURANT TABLE (4 Records)
INSERT INTO Restaurant (Restaurant_ID, Restaurant_name, Phone, Address, Rating) VALUES
(1, 'Truffles Gourmet', '08025531234', 'Koramangala 5th Block, Bengaluru', 4.7),
(2, 'Meghana Foods', '08025535678', 'Indiranagar 100ft Road, Bengaluru', 4.8),
(3, 'Empire Restaurant', '08025539999', 'Church Street, Bengaluru', 4.3),
(4, 'Pizza Bakery', '08025538888', 'HSR Layout Sector 1, Bengaluru', 4.5);

-- 3. POPULATING DELIVERY PARTNER TABLE (4 Records)
INSERT INTO Delivery_Partner (Partner_ID, Name, Phone, Vehicle_type, Partner_status) VALUES
(101, 'Ramesh Kumar', '9123456780', 'Motorcycle', 'Available'),
(102, 'Suresh Babu', '9234567812', 'Scooter', 'Busy'),
(104, 'Mahesh Gowda', '9456781234', 'Motorcycle', 'Offline');
```

[EXECUTION RESULT] Query OK, 13 rows affected (0.02 sec). All primary keys and check constraints verified.

3. SQL Data Insertion Script (Menu & Transaction Entities)

```
-- 4. POPULATING MENU_ITEM TABLE (8 Records linked to Restaurants)
INSERT INTO Menu_Item (Item_ID, Restaurant_ID, Item_name, Category, Price, Availability) VALUES
(201, 1, 'All American Burger', 'Main Course', 320.00, TRUE),
(202, 1, 'Peri Peri Fries', 'Appetizer', 140.00, TRUE),
(203, 2, 'Special Chicken Biryani', 'Main Course', 380.00, TRUE),
(204, 2, 'Paneer Butter Masala', 'Main Course', 290.00, TRUE),
(205, 3, 'Ghee Rice & Chicken Kebab', 'Combo', 310.00, TRUE),
(206, 3, 'Coin Parotta (Set of 3)', 'Breads', 60.00, TRUE),
(207, 4, 'Sourdough Pepperoni Pizza', 'Main Course', 550.00, TRUE),
(208, 4, 'Garlic Dough Balls', 'Appetizer', 180.00, FALSE);


-- 5. POPULATING ORDER_TABLE (5 Customer Orders)
INSERT INTO Order_Table (Order_ID, Customer_ID, Restaurant_ID, Order_Date, Total_amount, Order_status)
VALUES
(1001, 1, 1, '2026-10-24 12:30:00', 460.00, 'Delivered'),
(1002, 2, 2, '2026-10-24 13:15:00', 670.00, 'Delivered'),
(1003, 3, 3, '2026-10-24 19:45:00', 370.00, 'Preparing'),
(1004, 4, 4, '2026-10-24 20:10:00', 730.00, 'Pending'),
(1005, 5, 2, '2026-10-24 20:30:00', 380.00, 'Delivered');

-- 6. POPULATING ORDER_ITEM TABLE (Line Items for Orders)
INSERT INTO Order_Item (Order_item_ID, Order_ID, Item_ID, Quantity, Unit_price) VALUES
(501, 1001, 201, 1, 320.00),
(502, 1001, 202, 1, 140.00),
(503, 1002, 203, 1, 380.00),
(504, 1002, 204, 1, 290.00),
(505, 1003, 205, 1, 310.00),
(506, 1003, 206, 1, 60.00),
(507, 1004, 207, 1, 550.00),
(508, 1004, 208, 1, 180.00),
(509, 1005, 203, 1, 380.00);

-- 7. POPULATING PAYMENT TABLE (Payments linked 1:1 to Orders)
INSERT INTO Payment (Payment_Id, Order_ID, Payment_method, Payment_status, Payment_date, Amount) VALUES
(701, 1001, 'UPI', 'Successful', '2026-10-24 12:31:15', 460.00),
(702, 1002, 'Credit Card', 'Successful', '2026-10-24 13:16:00', 670.00),
(703, 1003, 'UPI', 'Successful', '2026-10-24 19:45:30', 370.00),
(704, 1004, 'Net Banking', 'Pending', '2026-10-24 20:10:05', 730.00),
(705, 1005, 'Cash', 'Successful', '2026-10-24 20:30:10', 380.00);

-- 8. POPULATING DELIVERY TABLE (Deliveries linked to Orders & Partners)
INSERT INTO Delivery (Delivery_ID, Order_ID, Partner_ID, Pickup_Time, Delivery_Time, Delivery_Status)
VALUES
(901, 1001, 101, '2026-10-24 12:40:00', '2026-10-24 13:02:00', 'Delivered'),
(902, 1002, 102, '2026-10-24 13:25:00', '2026-10-24 13:50:00', 'Delivered'),
(903, 1003, 103, '2026-10-24 19:55:00', NULL, 'Picked Up'),
(904, 1005, 101, '2026-10-24 20:40:00', '2026-10-24 21:05:00', 'Delivered');
```

[SYSTEM LOG] All 8 tables populated successfully. 0 integrity violations reported across 26 inserted records.

4. Verification Sample Query Output

Executing a verification join query across populated tables produces the active application state table below:

Order ID	Customer Name	Restaurant	Total (₹)	Order Status	Payment Method	Delivery Partner
1001	Aarav Sharma	Truffles Gourmet	460.00	Delivered	UPI	Ramesh Kumar
1002	Priya Patel	Meghana Foods	670.00	Delivered	Credit Card	Suresh Babu
1003	Rohan Verma	Empire Restaurant	370.00	Preparing	UPI	Ganesh Rao
1004	Sneha Reddy	Pizza Bakery	730.00	Pending	Net Banking	Unassigned
1005	Vikram Malhotra	Meghana Foods	380.00	Delivered	Cash	Ramesh Kumar